OPERATING SYSTEMS – ASSIGNMENT 1

TASK 11 – Recommend and Defend the Operating-System Plan (4 Marks)
CO1–CO5 | Bloom's Level: K5 – Evaluate

Introduction

The hospital needs an operating-system plan that supports safe patient monitoring, secure access to records, efficient medical-image processing and continuous database availability. A single OS configuration cannot satisfy every requirement equally well.

1. Recommended OS type and structure

Bedside monitors: Use a small embedded hard real-time OS, preferably with a microkernel or similarly fault-isolated design. It must be tested to meet the 10 ms alarm deadline.
Nurses' tablets: Use a handheld/mobile OS with power management, wireless support and device security.
Doctors' workstations: Use a desktop OS with user accounts, application support, endpoint protection and secure access to clinical systems.
Central database: Use a reliable server OS in a clustered configuration with fail-over, backups and replicated or shared storage.
MRI imaging server: Use a multiprocessor server OS, preferably supporting symmetric multiprocessing and high-speed I/O with DMA.
Partner hospitals: Use a distributed system with secure network communication and controlled data sharing.

2. Services, protection and I/O methods

Monitors need real-time scheduling, sensor I/O, alarm output, error detection and secure updates. Tablets need a touch interface, wireless communication, authentication and battery management. Workstations need application execution, file and network services, access control and audit logging. The database needs storage management, concurrency control, backup, recovery, authentication, authorisation and monitoring. The MRI server needs parallel processing, high-throughput storage and DMA. Network links must use encryption, authentication and access controls. Memory protection and process isolation should be enabled across all systems.

3. Limitations and risks

A real-time OS may support fewer general-purpose applications and requires careful testing. Mobile devices can be lost or stolen, so encryption, screen locks and remote management are important. Desktop systems need regular security updates. A cluster costs more and requires skilled administration; fail-over must be tested. A distributed system depends on network availability and consistent data exchange. DMA and high-speed imaging systems require compatible hardware and drivers. No design eliminates all failures, so monitoring, backups, incident response and recovery plans are necessary.

4. Suitability for three hospitals

The plan can support three hospitals by connecting their systems through secure distributed services. Each hospital can operate local monitoring and clinical systems while sharing authorised records through secure interfaces. The database design should support replication, backup and controlled synchronisation. Access should follow user roles and patient privacy requirements. Network failures should not stop local bedside alarms because the monitors must operate locally and independently.

5. Alternative concept: virtual machines on two large servers

Running every environment as virtual machines on two large servers may simplify central administration and resource allocation. However, bedside monitors should not depend on general-purpose virtual machines hosted in a central data centre because network delays, host scheduling and virtualisation overhead may affect the strict 10 ms alarm deadline. Local real-time devices are safer for this function.
Virtualisation uses extra memory for guest operating systems and hypervisor overhead. Multiple VMs also compete for CPU, memory and I/O resources. If one physical host fails, VMs on that host may stop; a second host can support recovery only if the system is configured with suitable redundancy and shared or replicated storage. Two servers also create possible shared bottlenecks and are not automatically fault-proof.

6. Final recommendation

Use separate, purpose-selected systems: embedded real-time OS for bedside monitors; mobile OS for tablets; desktop OS for workstations; clustered server OS for the patient database; and a multiprocessor OS for MRI processing. Connect hospitals through secure distributed services. Virtualisation may be used for suitable non-real-time server workloads, but patient-monitor alarm handling should remain local and independently reliable.

Conclusion

A mixed operating-system design best meets the hospital’s different needs. It prioritises patient safety, availability, security and performance while allowing the three hospitals to share records securely.

Prepared from the corresponding task in the uploaded CSA04 Operating Systems Assignment 1 document. Add your name, register number, required screenshots and your own measured evidence wherever requested before submission.